

BIOMECHANICAL ANALYSIS OF STANDING LONG JUMP FROM VARYING STARTING POSITIONS

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ABSTRACT

Mackala, K, Stodółka, J, Siemieniński, A, and Čoh, M. Biomechanical analysis of standing long jump from varying starting positions. *J Strength Cond Res* 27(10): 2674–2684, 2013—The purpose of this study was (a) to investigate the effect of the different foot movement (placement) during take-off and the initial knee joint angle used in standing long jump by the ground reaction forces analysis and 3-dimensional motion analysis (BTS SMART motion) and (b) investigate how the jump performances of different foot placement is related to the electromyography (EMG) activity (Noraxon) of 3 selected muscle groups (m. gastrocnemius, m. gluteus maximus, m. rectus femoris, m. tibialis anterior, m. biceps femoris, and m. vastus medialis). Six high caliber sprinters (100 m: 10.87 ± 0.38 seconds and 400 m: 46.75 ± 1.05 seconds) performed a series of jumps from parallel and straddle foot placement at take-off on a 2 force platform (Kistler model 9286B) to determine if a different pattern of take-off improves jumping distance. Using kinematic and kinetic data, the knee joint angle, the trajectories of center of mass (COM), magnitude of take-off peak force, and impulse during take-off phase were calculated. Average standing long jump performances with straddle foot placement were 13.58 cm (5.18%) above that from parallel feet placement. The take-off velocity with 90° knee initial angle initiation of take-off was not different (1.18 and 1.17 m·s⁻¹, respectively) between the 2 jumps. The take-off angles on the COM trajectory also showed differences (69.87 and 66.8°, respectively) between each other. The contribution (EMG activation) made by the 6 muscles were almost the same during all phases for the 2 jumps; however, some differences can be found, in either unilateral (single leg) or sums of both legs (bilateral) measurements. A recommendation can be formulated that the contribution of straddle foot placement during take-off can significantly

increase the value of power measurement especially when the evaluation requires a complex movement structure with the division on the left and right legs, for example, sprint start from block.

KEY WORDS horizontal jump, straddle position, parallel position, countermovement

INTRODUCTION

The standing long jump is considered a fundamental motor skill for a variety of sports where high-velocity contractions are demanded: sprinting, hurdling and jumping in athletics, football, ski jumping, and some combat sports. It is also often used as one of the best functional tests to assess explosive power of the lower extremity.

The performance (length of jump) is highly dependent on numerous independent and dependent variables. One is the application of proper jumping technique—motor proficiency, which may influence the achieved results and underestimate the level of explosive power of lower extremities. Lee and Cheng (12) indicate that the performance of standing long jump may be affected by level of countermovement, maximum joint and muscular strength, and starting posture. Cheng and Chen (5) also claimed that jump distances are insensitive to the starting position, which is determined by angle of knee flexion and posterior position of the trunk at take-off. Successful performance in the standing long jump depends also on a high level of coordination of both the upper and lower body segments (2, 15).

Because horizontal jumping is a movement that requires complex motor coordination (Wakai and Linthorne [19]) of both upper and lower body segments, many studies have investigated the role of arm motion (swing of upper extremities) in standing long jump performance (1,3,10). The double forward swing of the arms generates the angular momentum that is transferred to the body as a whole (16). According to Ashby and Heegaard (3), the better results in the standing long jump (by ~21%) were achieved when the free arm movement was allowed. This is quite substantial compared to vertical jumps, where the arm swing contributes only 8.6–8.8% of its height (5). Those authors claimed that arm movements can be used to regain or maintain

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TABLE 1. Basic morphological characteristics and level of sprint performance of the tested sprinters ($n = 6$).

	Body parameters						Best results	
							100 m (s)	400 m (s)
	Age (y)	Height (cm)	Weight (kg)	Leg length (cm)*	Body mass index (<i>n</i> = 6)	Rohrer (<i>n</i> = 6)	(<i>n</i> = 3)	(<i>n</i> = 3)
<i>X</i>	21.6	186.41	78.16	98.35	22.53	1.20	10.87	46.75
<i>SD</i>	2.73	5.13	8.15	3.42	2.31	0.13	0.38	1.05

*Leg length was measured from the greater trochanter on the femur to the sole of the foot (medial malleolus).

balance through the transfer of angular momentum to the arms from the rest of the body.

No one has attempted to determine the effects of various foot setting position on the quality of standing long jump; therefore, the role of parallel or straddle foot placement to the starting line at the instant of take-off in this activity remains unclear. We predict that better distances can be attained in the standing long jump from straddle position. This presumption does not arise only from the difference in foot setting, but above all also in differences in take-off characteristics, the knee joint angle, the trajectories of center of mass (COM), magnitude of the take-off peak force. These differences can be used as motor proficiency discriminators when evaluation requires other complex movement structures; broken down into a single movement or the movement of both limbs, for example, sprint start from block or start in swimming.

The purpose of this study was to investigate the (a) effect of the different foot movement (placement) throughout the countermovement, take-off phase and initial flight phase. More specifically, it was of interest to determine to what extent the initial knee joint angle, the ground reaction forces, and velocity at take-off, had on improving the length of the jump. (b) investigate how the jump performances of different foot placement related to electromyography (EMG) activity of 3 selected muscle groups (m. gastrocnemius, m. gluteus maximus, m. rectus femoris, m. tibialis anterior, m. biceps femoris, m. vastus medialis). Therefore, to better understand this problem, we need to ask a few support questions: (a) Can standing long jump performance (the greatest jump distance possible) be improved when jumping with straddle foot placement? (b) Which parameters most determine the performance of the jump from parallel and straddle positions? (c) Which value of the mechanical parameters of the jumping motion may be the most relevant metrics (motor proficiency discriminators)?

Based on these questions, we feel that the identification of any differences in the kinematic, dynamic, and EMG data can bring valuable information for practitioners on

usefulness of horizontal jumps for lower extremity power evaluation or determinant of the level of motor proficiency.

METHODS

Experimental Approach to the Problem

During the experiment, each subject performed 3 maximum effort trials of 2 standing long jump; one from parallel (wide apart) feet placement behind the starting line at take-off, second from self-selected straddle position. The order in which the 2 conditions of jump was performed was standardized. (a) Parallel standing long jump—from the erect position, the feet are positioned shoulder width apart or slightly more, parallel to the starting line, than subject dynamically lowers the center of body mass by flexing the knee to about 85–90° (countermovement). Next, by application of free arm motion (double arm swing during entire jump) and knee extension, the subject executes the jump as far as possible. (b) Straddle standing long jump—from the erect position, the feet are positioned in self-selected straddle position (about 30–40 cm apart and with the left foot in front), then the subject dynamically lowers the body COM by flexing the knees to about 85–90° (countermovement). Next, by application of free arm motion (double arm swing during entire jump) and knee extension, the subject executes the jump as far as possible.

All the jumps were performed in the biomechanical laboratory on a specially constructed wooden floor on which 2 force plates (Kistler 9286B) were installed. The subjects stood on the platforms in such a way that each leg was placed on a separate platform and on the command “ready” performed a maximal horizontal jump. The landing was executed on a special exercise mat. It was emphasized to the subjects the importance of simultaneous take-off from 2 legs and proper amortization when landing. No foot movements before take-off were allowed. The experiment was conducted in June, that is, during competition season, and, consequently, the sprinters were at a high performance level. Tests were arranged in a weekly training microcycle to correspond with the date on which athletes should perform dynamic strength training and jumping. The study was

TABLE 2. Mean values and SD ($n = 18$) for the starting position, ground reaction forces, kinematic, kinetic and muscle activation variables for standing long jump from different feet placements ($p < 0.05$).

Variables	Standing long jump			
	Feet placed parallel		Feet placed in straddle	
Starting position				
Feet parallel to starting line	38 (0.03)			
width—right/left big toe (cm)				
Width—right/left ankle bone (cm)	29 (0.02)			
Feet in straddle position				
Length—right/left big toe (cm)	28 (0.02)			
Length—right/left ankle bone (cm)	24 (0.02)			
Width—right/left big toe (cm)	26 (0.02)			
Width—right/left ankle bone (cm)	17 (0.02)			
Ground reaction force variables	Right leg	Left leg	Right leg	Left leg
Peak vertical ground reaction force (% body weight) (N)	118.12 (15.29)	126.65 (33.24)	133.27* (22.09)	119.30 (23.25)
Peak horizontal ground reaction force (% body weight) (N)	48.91 (13.54)	42.77 (12.83)	57.07* (13.40)	41.72 (18.95)
Peak force (% body weight) (N)	180.71 (28.23)	180.73* (49.51)	197.31* (41.90)	168.77 (48.51)
Center of gravity motion variables				
Take-off height (m)	0.92 (0.06)		0.90 (0.08)	
Jump length (m)	253.94 (9.89)		267.11* (13.52)	
Time of flight phase	0.662 (0.017)		0.660 (0.042)	
	0.661 (0.025)		0.629 (0.038)	
Projection angle (°)	69.87* (16.04)		66.84 (16.40)	
Velocity of center of gravity projection ($m \cdot s^{-1}$)	1.18 (0.44)		1.17 (0.59)	
Vertical acceleration of center of gravity ($m \cdot s^{-2}$)	2.98 (5.4)		3.31* (3.679)	
Kinematic variables	Right leg	Left leg	Right leg	Left leg
Hip angle at take-off (°)	9.00 (12.34)	10.96 (12.32)	4.06 (8.899)	9.93 (19.39)
Knee angle at take-off (°)	13.44 (11.39)	15.13 (10.77)	14.98 (7.48)	13.32 (19.30)
Ankle angle at take-off (°)	-24.79 (10.04)	-29.00 (10.77)	-25.16 (10.59)	-30.35 (10.30)
Peak hip flexion angle (°)	72.70 (29.36)	79.81 (37.23)	75.85 (21.49)	74.53 (28.70)
Peak knee flexion angle (°)	80.79 (22.66)	82.74 (19.98)	81.15 (8.37)	82.02 (12.89)
Peak ankle flexion angle (°)	29.39 (7.14)	26.65 (11.75)	32.34 (9.13)	26.67 (9.71)
Peak trunk flexion angle (°)	62.23 (23.44)	64.90 (16.83)	72.67* (26.29)	57.53 (9.15)
Kinetics variables	Right leg	Left leg	Right leg	Left leg
Peak hip joint moment ($N \cdot m$)	2.15 (0.85)	2.50 (1.14)	2.67* (0.83)	2.68* (0.034)
Peak knee joint moment ($N \cdot m$)	1.09 (0.47)	1.19 (0.24)	1.32* (0.56)	0.91 (0.30)
Peak ankle joint moment ($N \cdot m$)	1.59 (0.14)	2.06* (0.22)	1.83 (19)	1.87 (0.030)
Hip joint power ($W \cdot kg^{-1}$)	9.04* (3.89)	6.33 (2.93)	5.98 (4.12)	12.17* (4.91)
Knee joint power ($W \cdot kg^{-1}$)	4.04* (3.93)	1.92 (4.42)	1.07 (3.41)	1.89 (2.27)
Ankle joint power ($W \cdot kg^{-1}$)	2.32* (2.39)	2.65 (94.45)	0.59 (2.11)	2.74 (1.80)
Muscle activation at take-off (% MVC)	Right leg	Left leg	Right leg	Left leg
Tibialis anterior	42.33 (19.30)	33.31 (27.89)	43.47 (15.34)	31.49 (28.34)
Gastrocnemius	145.31* (48.32)	124.73 (34.38)	136.43 (52.57)	139.08* (35.14)
Biceps femoris	26.84 (13.28)	51* 85 (52.58)	40.15 (938.33)	47.26 (939.14)
Vastus medialis	149.28* (81.49)	105.70* (35.20)	92.42 (60.87)	60.58 (37.83)
Rectus femoris	94.84* (37.04)	57.70* (928.50)	63.74 (51.52)	38.47 (24.98)
Gluteus maximus	199.70* (98.70)	99.39* (61.53)	169.26 (75.00)	48.76 (24.15)

*A difference between the groups is statistically significant ($p < 0.05$). %MVC, Maximum Voluntary Contraction measured in %.

conducted in the morning, which corresponded to the daily training time of the sprinters. The subjects were instructed on how to perform the 2 different jumps, to be well fed, rested, and adequately hydrated before the testing days.

Before testing, the subjects were allowed to perform 10–15 minutes individual warm-up, including light jogging, stretching, and light form of jumps exercises. Considering the reliability, the athletes did not bring more practice to

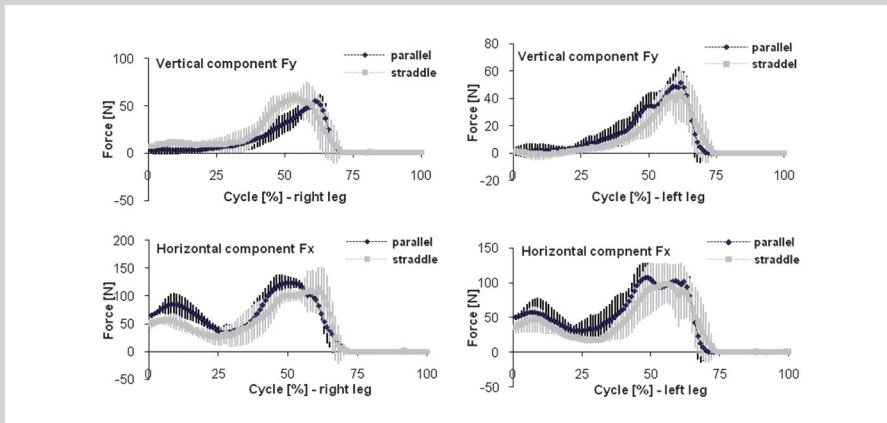


Figure 1. Comparison of the peak values of the horizontal and vertical ground reaction forces shown separately for the left and right lower extremities.

the test before the experiment in any form of jumping. The result is that sprinters are very familiar with the jumping technique from the parallel position. In turn, during the jumping from the straddle position, all the sprinters placed their left limb in the front, similar to the position which they use in the sprint start from the block. Because for them it is a natural placement, they did not require additional practice. All trials of each subject were included in the data analysis.

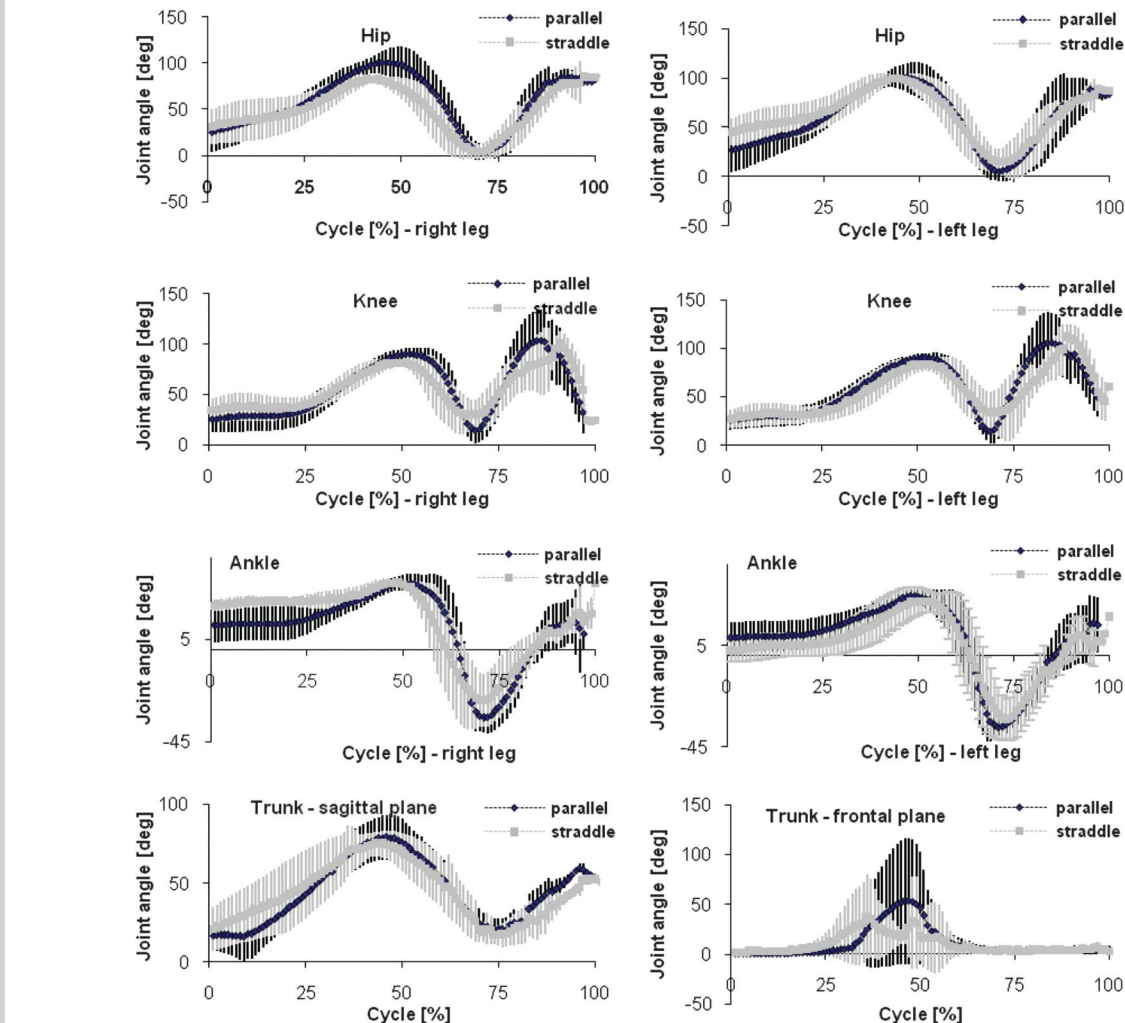


Figure 2. The illustration of the main joint angles (ankle, knee, hip, pelvis, and trunk).

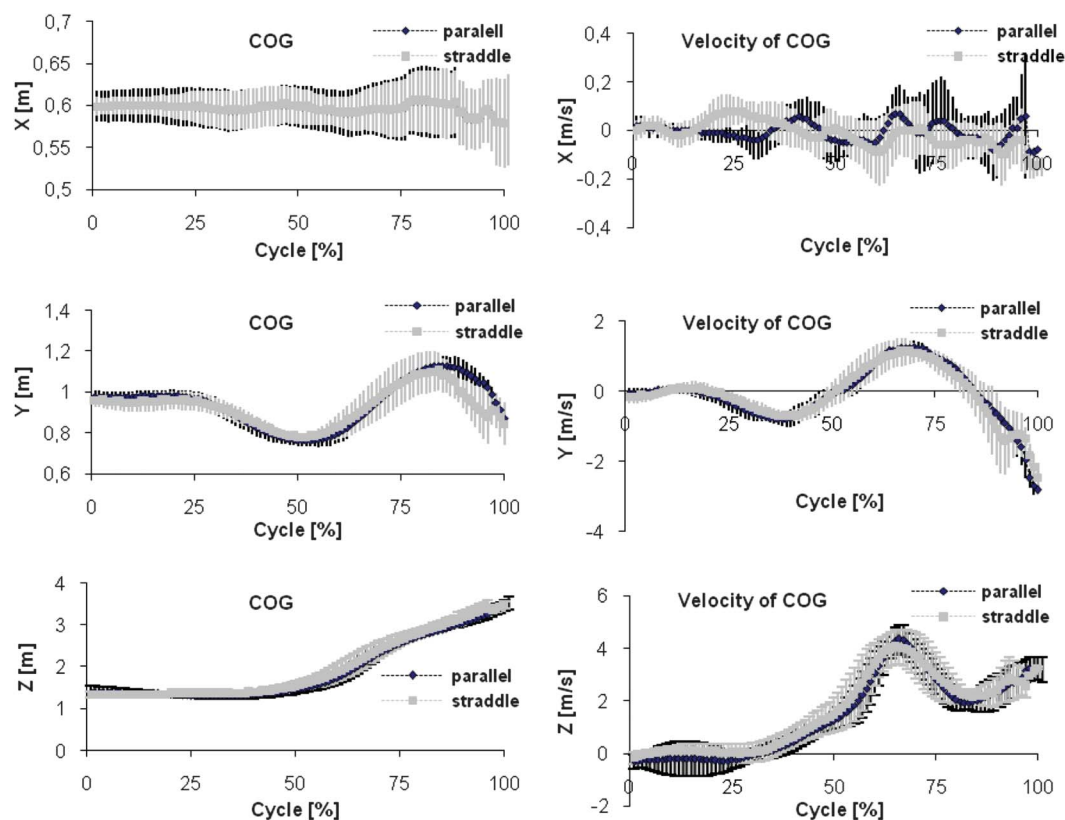


Figure 3. Illustration of the trunk segment in the vertical, horizontal, and side positions and the value of the speed of center of gravity.

Subjects

Six male high caliber sprinters (100 and 400 m) volunteered to participate in the experiment (Table 1) (age = 21.6 ± 2.73 years, height = 186.41 ± 5.13 cm, weight = 78.16 ± 8.15 kg, 100-m performance = 10.87 ± 0.38 seconds and the best results = 10.54 seconds, 400-m performance = 46.75 ± 1.05 seconds and the best results = 45.54 seconds). Before providing their written informed consent for participation, all procedures and testing protocols were clearly explained to the subjects. The study was approved by the Human Ethics Committee of the University School of Physical Education in Wrocław.

Measurement of Jumping Ground Reaction Forces

Ground reaction forces were obtained by using 2 Kistler (Type 9286B, Kistler Instrumente AG, Winterthur, Switzerland) force plates, independently set up. The sampling frequency was 240 Hz. This accurately recorded take-off and landing times to assess the duration of the flight phase and hence calculated the jump height. The measurements were computer recorded to give a numerical and graphical picture of the 3 components of the ground reaction forces (F_x , F_y , and F_z).

The 3 components of moments of forces and the resultant force vector (lengths, points of application and inclination angles of these vectors). Ground reaction forces were measured using 4 piezoelectric sensors placed on the corners of the platform, linked directly to the amplifier, which connects via an analog-digital converter with the computer. BioWare program provided by the manufacturer of the platform was used to collect and analyze data. With the graphical presentation of the jump, it was possible to read the maximum values of forces during the take-off and landing. Thanks to the 2 dynamometric platforms, it was possible to register the peak ground reaction forces unilaterally and bilaterally. The force has further been normalized according to the percentage of body weight (BW) of measured subjects (Newtons).

Measurement of Jumping Kinematics Variables

The kinematic measurement of the vertical jumps and standing long jumps involved the following variables: take-off angles in the ankle, knee and hip joints, take-off height, the flight distance of the body, flight time, total distance, and take-off. They were measured using the video recording system (BTS Smart-E), which includes infrared cameras

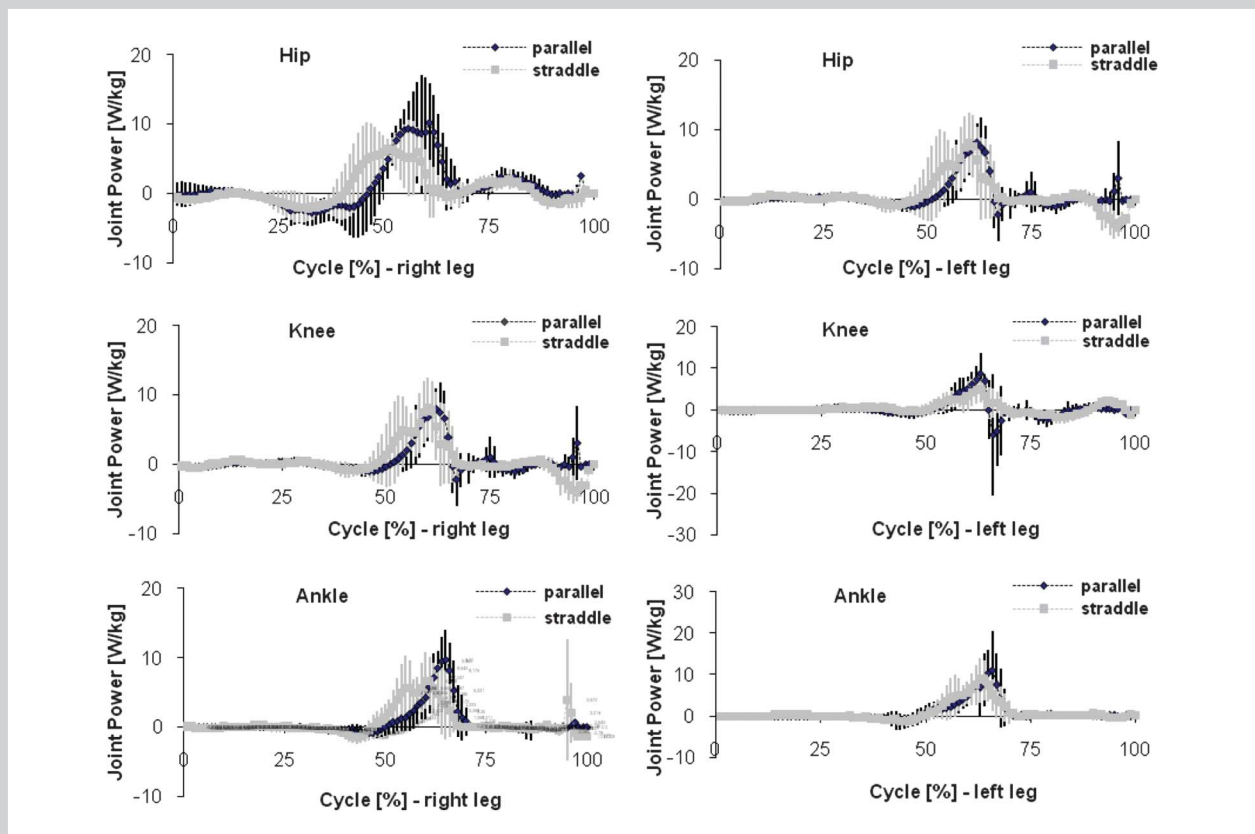


Figure 4. Illustration of the hip, knee, and ankle joint power (watts per kilogram).

TVC in the 6-variable-12 mm “zoom,” 120 Hz, enabling high positioning accuracy of the identification markers: <0.2 mm over $3 \times 2 \times 2$ m. Each camera was set such that the optical axis perpendicular to the lens passes through the center of the start, the first 3 steps along the catwalk, starting place of the vertical jump and the standing long jump landing. During the exercise, the subject will be positioned in both optical axes in the frontal plane, and sagittal. To avoid errors resulting from the inertia of the mechanism of the camera, the device was turned on 5 seconds before the start of each exercise. To determine the angles of body segments and the position of the center of gravity, to develop a kinematic model, a system of 19 markers was used. The retroreflective markers were firmly attached at the following locations on the right and left sides of the body: head, shoulders, forearm, upper arm, torso, hips, thigh, calf, and foot (18).

Measurement of Electromyography Muscles Activation

The EMG activity—signals from 6 muscles of the left lower extremity and 6 of the right extremity (m. gastrocnemius, m. gluteus maximus, m. rectus femoris, m. tibialis anterior, m. biceps femoris, m. vastus medialis) were recorded telemetrically. A 16-channel EMG device (Noraxon TeleMyo2400 G2)

was used. It consisted of 2 units: a mobile unit captured all EMG signals and transmitted them to the stationary unit using wireless technology (Wi-Fi). Surface EMG muscle activity was detected by means of bipolar surface electrodes Ag-AgCl (Noraxon), which were glued to the specific location on the belly of each muscle, after through skin preparation. The electrodes were fastened by qualified medical staff. The recorded signals were filtered and smoothed. First, a high-pass filter was used—filtration of low-band/low-pass/Butterworth/300 Hz to eliminate the artifacts. Then a low-pass filter filtration of high-band/Pass/Butterworth/10 Hz. frequency was used. Next, the root mean square value of the EMG signal was evaluated in a 0.1-second long moving window. The results are presented with the duration of the task scaled to equal 100%. The EMG activities were full wave rectified, time normalized and integrated (percentage) into phases. From the 0 to about 55% of the cycle, we have a phase known as the countermovement phase, in which there is a dynamic work of the upper limbs, and the largest flexion at the knees. Then begins the so-called push-off—the initiation of the take-off movement. This phase lasts for about 70% of the entire cycle. This is followed by detachment of the feet from the ground—the end of the propulsion phase and the beginning of the flight phase.

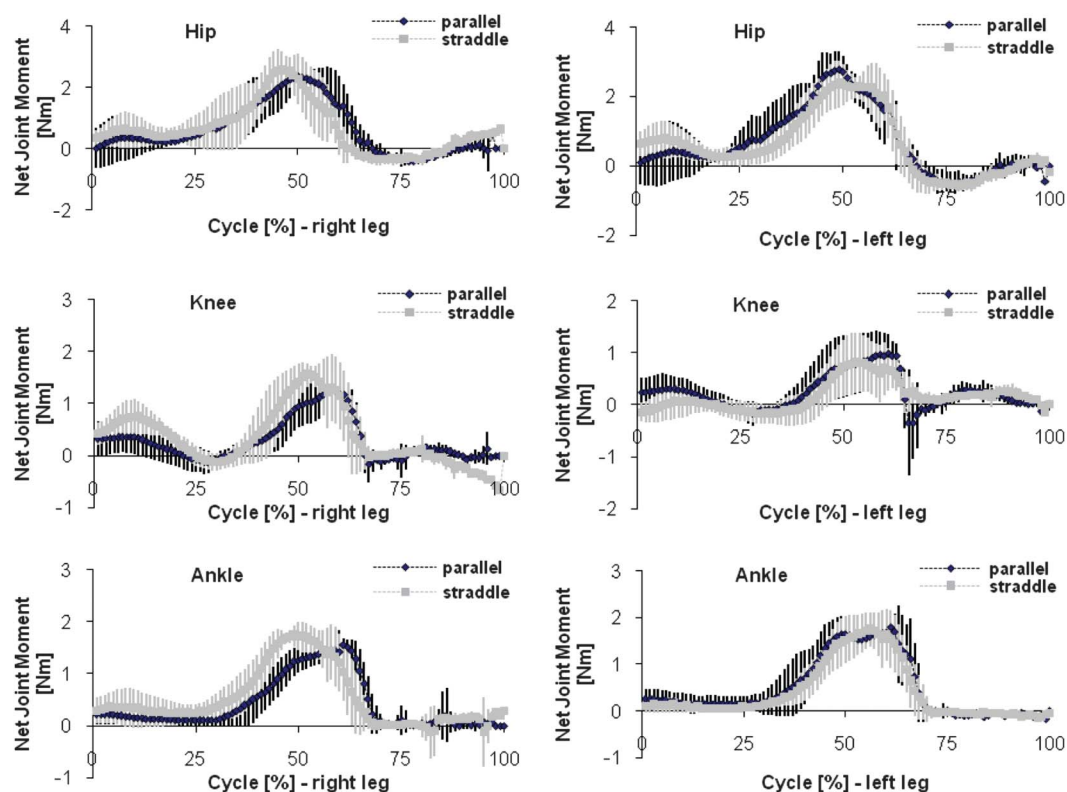


Figure 5. Illustration of the hip, knee, and ankle net joint moment (newton meter).

Statistical Analyses

Descriptive statistics included mean $\pm$ *SD*. Statistical significance was set up at $p \leq 0.05$. After establishing a significant main or interaction effect, the data were evaluated with a least significant difference post hoc analysis for differences among means at the $p \leq 0.05$ level.

RESULTS

The subjects jumped 5.18% further ($p < 0.05$) with average distance of 267.11 ± 13.52 cm from straddle feet placement at take-off and 253.94 ± 9.89 cm from parallel foot position (see Table 2 for a summary of results). The jump is measured as the horizontal distance from the take-off line to the closest mark made by the heels at landing.

In the vertical and horizontal force-time curves, we found that there are no significant differences between the parallel foot placement and the straddle feet placement in the duration of force application (Table 2). The peak force value was observed during the end of push-off and before the instant of take-off, which represents about 65–70% of the entire jump cycle. A bigger peak force (% BW) value is observed in the vertical component of the standing long jump executed from the straddle position with special attention toward the left leg, 197.31 N (Figure 1).

The hip, knee, and ankle joint angles were almost identical between these 2 jumps before the instant of take-off (Table 2, Figure 2); however, as we supposed, the bigger knee joint angle during the initiation of the push-off (beginning of the take-off phase) represents the jump executed from the straddle position, respectively: left leg 91.36° and right leg 91.77° compared with the straddle position (left leg 85.87° and right leg 88.58°).

The magnitude of hip joint flexion was greater in the jump from the straddle position than in the parallel foot placement, both left and right lower extremities. It was similar with the peak trunk flexion. The angular displacements of the COG were remarkably similar between both jumps during almost the entire jump, the horizontal, vertical, and side positions of the trunk (Figure 3). There were no significant differences in the capacity of activation, although some muscles showed some differences. Particularly evident is the gluteus maximus muscle activity and anterior tibialis (Table 2). This applies to the right leg in the jump from the straddle position.

Joint power (watts per kilogram) between the 2 jumps shows big differences especially when the 2 lower extremities are analyzed separately (Table 2, Figure 4). The left extremities showed a higher value in all joints (hip 12.17, knee 1.89, and ankle 2.74) in the jump from the straddle

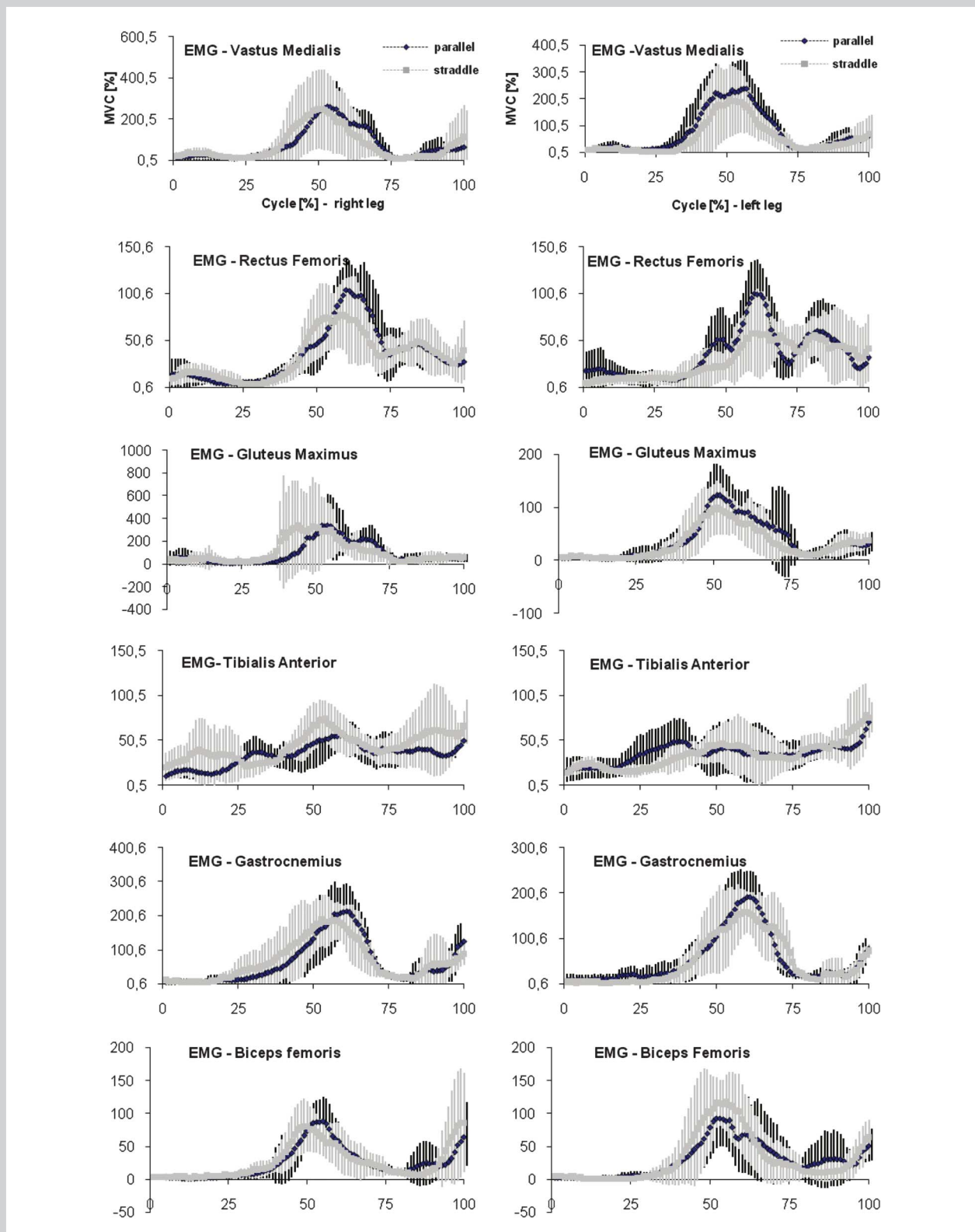


Figure 6. Illustration of the selected muscles activities electromyography in the standing long jump.

position compared with the right ones. The opposite relationship appeared in the jump from the parallel arrangement of the feet. Here, larger values were reached in the right limb with the exception of the ankle joint, where the values were almost equal (2.32 and 2.58, respectively).

DISCUSSION

This study is the first to describe the differences in the kinematics and kinetics of the standing long jumps performed; from the parallel foot placement and from the straddle at the instant of take-off position. More specifically, it was of interest to determine to what extent the initial knee joint angle, the ground reaction forces and velocity at take-off do to improve the length of the jump. The second purpose was to investigate how the jump performances of different foot placement related to EMG activity of selected 6 muscles (m. gastrocnemius, m. gluteus maximus, m. rectus femoris, m. tibialis anterior, m. biceps femoris, m. vastus medialis).

The standing long jump (broad jump) was chosen because it (and its derivatives) is commonly used in the athletic training in various sports. Furthermore, this exercise or test represents an explosive type of movement, which correlates well with other types of explosive movements such as the vertical jump and sprinting (11,14,17,20). However, in the past, the comparisons focused only vertical and horizontal jumps (9,10,13) where feet were placed parallel to the starting line.

In this study, the question that straddle positioning of the feet during take-off can improve performance in standing long jumps was conclusively confirmed. Better results (distance attained) were reached in the standing long jump from the straddle position by about 13.58 cm. From the practical stand point, this is quite a significant difference.

Both jumps are referred to as countermovement jumps where the double arm swing is performed. In our protocol, the application of the upper extremity movement in the entire jump was necessary to contribute more to the efficiency (distance) of the jump and to better coordinate the body segment movement, especially the lower extremities. The double forward swing of the arms generates the angular momentum that is transferred to the body as a whole (16). Ashby and Heegaard (3) stated that there was about a 21% improvement in the standing long jump when free arm movement was allowed. This is quite substantial compared with that of vertical jumps, where the arm swing contributes only 8.6–8.8% of its height (5). Because no one had investigated the effect of the foot setting on the standing long jump length, we have no reference point. However, a better result obtained in the long jump with a straddle position (increase by 5.18%) compared with the jump improvement where free arm movement was applied seems small. The explanation for such difference may lie first of all in the abilities to transfer jumping technique from the parallel position to straddle and the experience of sprinters and coordination skills.

According to Wu et al. (21), the most important elements that promote standing long jump performance are the ability to coordinate the arm movement and selection of the appropriate take-off angle. The selection of the take-off angle is closely related to the direction of the jump. The angle of projection (trunk at the beginning of take-off) obtained in the jump from the parallel foot placement was 69.97°, whereas this parameter obtained in the jump from the straddle position was 66.84° (Figure 3). The explanation of this discrepancy can be related to the knee joint angle during beginning of the take-off phase. The bigger knee joint angle at the initiation of take-off (Figure 2) showed the jump executed from the straddle position (left leg 91.36° and right leg 91.77°) compared with the straddle position (left leg 85.87° and right leg 88.58°). By combining these 2 variables, we can confirm that the whole body was tilted more in the forward direction in the straddle foot placement position than in the parallel. Therefore, the lower trunk flexion angle at the hip and lower angle of the knee in the jump from a straddle position generates a larger momentum in both forward and upward movements during take-off; this translates into a better performance. On the other hand, the jump from parallel position required a straighter posture and less knee angle in order to keep the body's COM more over the feet; this can generate more upward momentum than in the jump from the straddle position. This result is not consistent with the other findings because of lack of data; however, Fukushima et al. (9) observed that horizontal jumps require a bigger forward trunk movement, which is automatically associated with a greater angle during take-off compared with vertical jumps. They referred to this as a rotation-extension strategy. During the jump, the rotation of the body mass forward in relation to the base of support must take place before the moment of leg extension in the ankle and knee joints. A main criterion of an efficient transformation of rotational movement into translatory movement in both type jumps; vertical and horizontal, is the velocity of body's center of gravity at the time of take-off. In this parameter, no differences have been noticed between the 2 jumps (1.17 and 1.18 milliseconds⁻¹, respectively). This means that improvement in standing long jump performance from the straddle position was not because of the increase of the CG velocity at the take-off, leading to the conclusion that both jumps equally use the proximal-distal principle of the muscular chain in an optimal way. This information confirms what Ashby and Heegaard (3) claimed that the majority (71%) of the improvement in standing long jump performance with a free arm application results from the increase in the CG velocity at take-off.

Standing long jumps are typical examples of bilateral activity. Nevertheless, differences between these 2 activities exist particularly from the aspect of ground reaction forces. In the phase of initiation of take-off, the vertical ground reaction (% BW) force in the parallel foot placement jump

amounts from 118.12 to 126.65 N on right and left legs, respectively. The sum of ground reaction force on both legs is thus 244.77 N (Table 2, Figure 5). In straddle foot placement jump, the vertical ground reaction (% BW) force achieve in average a bilateral ground reaction force 252.57 N. Unilateral ground reaction force amounts to 133.27 N for the right leg and 119.30 N for the left. Similarly, the sum of peak joint moments (newtons per meter) of the jump is on average higher in the straddle foot placement position, compared with the parallel foot (5.35 and 4.65 Nm, respectively). According to these data, standing long jump from the straddle position with a higher ground reaction force was capable of realizing a longer jump, than the jump from a parallel foot position. No data existed in the literature to compare these findings; however, some comparisons can be made. In the phase of maximal sprinting velocity, the vertical ground reaction force amounts to 1,300–1,600 N on each leg. The sum of ground reaction forces on both legs is thus between 2,600 and 3,200 N (14).

When examining the muscle activation (Figure 6) profiles in both jumps, it can be observed that the highest activation showed the flexor muscles (m. rectus femoris and m. tibialis anterior) during the countermovement phase (greater knee joint flexion motion) in the parallel foot placement position. Contrary to this during the push-off phase, the greater activation showed the hip joint extensors muscle such as m. gluteus maximus and m. biceps femoris in the straddle foot placement. In turn, when we take into account muscle activations divided into left and right limbs, the magnitude of muscles activation measured on 2 separate force platforms differ substantially. The strongest activation showed the muscles from left limb. This may suggest that muscles from left limb are adapting more or less toward the joints moments when jumping from different foot placement at the moment of push-off. This may be because of differences in the single limb angle when jumping from straddle position compared to the parallel foot placement when the angles are equal for both limbs. This was confirmed by Brockett et al. (4) research where the m. gluteus maximus is responsible for developing high force because of relatively low angular velocity of hips. They also claimed that the highest level of activation of the thigh muscles is at the beginning of hip extension. Similar reasoning is presented by Fukashiro et al. (9). When the take-off action continues, the knee joint extensor muscles mainly m. rectus femoris transfer the energy from hip into the knee. The last phase of take-off is concluded with the biarticular calf muscles (m. gastrocnemius). Because the jump directions do not differ substantially between the 2 jumps, the powerful execution of jumps depends mainly on the single and double-joint muscle coordination. These biarticular muscle activations are reviewed in previous studies (8,7,6).

In conclusion, statistically significant ($p < 0.05$) differences between the standing long jumps of both types: from parallel

and straddle foot placement at take-off were revealed in some kinematic and dynamic parameters. The differences between the jumps were revealed in the length of the jump (5.18%). The explanation for this difference may be the experience of sprinters, showing their coordination skills, and first of all their abilities to transfer technique from standing long jump from the parallel position to the straddle. Significant differences can also be found, either in unilateral (single leg) or sums of both legs (bilateral) of measurement of COM projection angle (degrees) vertical acceleration of body center of gravity, vertical and horizontal ground reaction forces (% BW), and the impulse of force in the concentric and eccentric phase of the jump. The lower limb extensor muscles (m. gluteus maximus and m. biceps femoris) exerting more force when they are in the straddle foot placement because of differences in the single limb angle compared to the parallel foot placement when the angles are equal for both limbs.

PRACTICAL APPLICATIONS

A few important implications for the practitioners may be derived from this investigation.

The horizontal standing long jump made from both parallel and straddle leg alignment during take-off are very important tools in explosive power training development for several sports. These 2 exercises can also be used to improve functioning of plyometric (eccentric-concentric muscular work) in lower extremities. Furthermore, these jumps are reliable and objective measuring instruments for diagnosing the power ability of lower extremities. However, the contribution of straddle foot placement during take-off can increase the value of lower extremities power measurement. It is necessary when the evaluation requires a complex movement structure with special reference to the division on the left and right leg (e.g., sprint start from block). Because of the relatively small sample of subjects measured, the results of the present experiment need to be considered with some degree of scientific caution, although some results are consistent with previous research on standing long jump.

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